

Heavy-Tailed Stochastic Optimization

$$\min_{x \in \mathcal{X}} F(x) := \mathbb{E}_{\xi \sim \mathcal{D}} [f(x, \xi)].$$

$$\text{Stochastic Gradient Descent: } x_{t+1} = \Pi_{\mathcal{X}}(x_t - \eta_t \nabla f(x_t, \xi_t)).$$

A.1. Finite p -th moment, i.e. $\exists G > 0$ such that $\mathbb{E} \nabla f(x, \xi) = \nabla F(x)$,
 $p \in (1, 2]$, $\mathbb{E} \|\nabla f(x, \xi)\|^p \leq G^p$.

Example: Two-sided Pareto distribution: $\Pr(|\xi| \geq s) \sim s^{-\alpha}$, $p < \alpha$.

Motivation. Case $p < 2$ empirically observed in image classification [SSG2019], LLMs [ZKV+2020], reinforcement learning [GZP+2021].

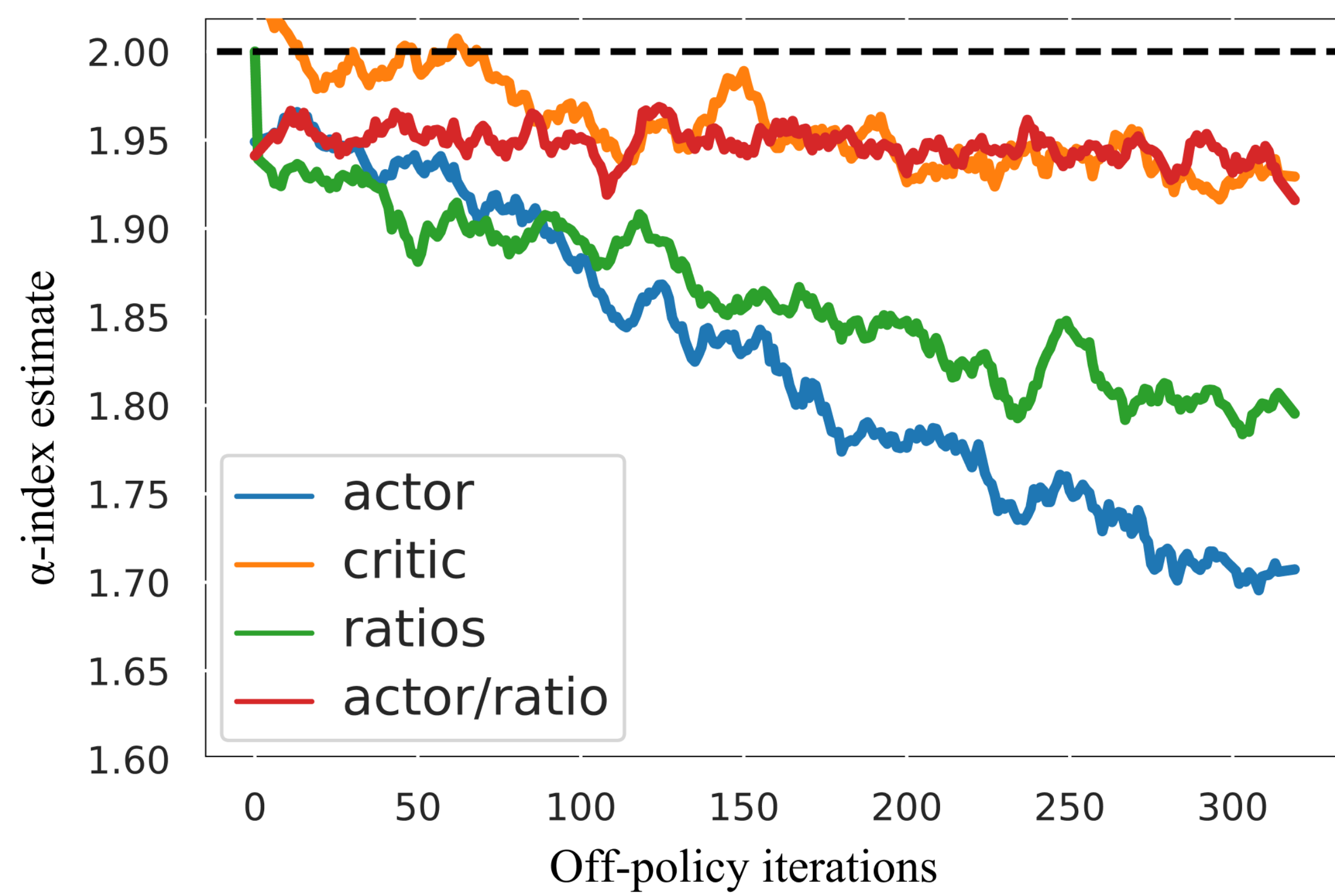


Figure 1: Heavy-tails in Proximal Policy Optimization (without clipping) during off-policy steps at various stages of training iterations in MuJoCo environments. Alpha index is an estimate of p aggregated over eight environments [GZP+2021].

Summary of SGD Step-sizes and Complexities

	Convex	Strongly Convex	Non-convex
Converg. $\leq \varepsilon$	$\mathbb{E} F(x) - F^*$	$(\mathbb{E} (F(x) - F^*)^{p/2})^{2/p}$	$\mathbb{E} \ \nabla F(x)\ ^2$
Step-size	$\eta_t = \frac{D_{\mathcal{X}}}{G t^{1/p}}$	$\eta_t = \frac{2}{\mu t}$	$\eta_t = \left(\frac{\Delta_1}{L_p G^p t} \right)^{1/p}$
Compl. in $\Theta(\cdot)$	$\left(\frac{GD_{\mathcal{X}}}{\varepsilon} \right)^{\frac{p}{p-1}}$	$\left(\frac{G^2}{\mu \varepsilon} \right)^{\frac{p}{2(p-1)}}$	$\Delta_1 \left(\frac{L_p^{1/p} G}{\varepsilon^2} \right)^{\frac{p}{p-1}}$

Advantages of Vanilla SGD:

- ✓ Simple, no extra parameters to tune.
- ✓ Parameter-free (wrt. p) in strongly convex case.
- ✓ Implicit model-compressibility.

Challenge of Infinite Variance

Let $F(x) = \frac{1}{2}x^2$ on $\mathcal{X} = \mathbb{R}$, ξ is Pareto with $\alpha \leq 2$. Then

$$\mathbb{E} [F(x_2)] = \frac{1}{2} \mathbb{E} [\|\nabla F(x_2)\|^2] = \frac{1}{2} \mathbb{E} [\eta_1^2 \|\xi_1\|^2] = +\infty.$$

No convergence in expectation! Why and when does SGD work?

Main Contribution

A.2. (L_ν, ν) -Hölder smooth. Let $\nu \in [1, 2]$, $L_\nu > 0$.

$$|F(x) - F(y) - \langle \nabla F(y), x - y \rangle| \leq \frac{L_\nu}{\nu} \|x - y\|^\nu.$$

Upper and Lower Bound for SGD

Theorem 1. Under **A.1.**: 1. If $F(\cdot)$ is convex, $\|x - x^*\| \leq D_{\mathcal{X}}$:

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E} F(x_t) - F(x^*) \asymp \frac{GD_{\mathcal{X}}^{2-p} \|x_1 - x^*\|^{p-1}}{T^{\frac{p-1}{p}}}.$$

2. If $F(\cdot)$ is strongly convex, $\bar{x}_T \sim \mathcal{U}\{x_1, \dots, x_T\}$:

$$\mathbb{E} (F(\bar{x}_T) - F(x^*))^{p/2} \asymp \frac{G^p}{\mu^{p/2} T^{p-1}}.$$

3. If $F(\cdot)$ is non-convex, **A.2.** with $\nu = p$, $\Delta_1 := F(x_1) - F^*$:

$$\mathbb{E} \|\nabla F(\bar{x}_T)\|^2 \asymp \frac{L_p^{1/p} \Delta_1^{\frac{p-1}{p}} G}{T^{\frac{p-1}{p}}}, \quad \text{see Fig. 2. for lower bound.}$$

Key analysis tool: Hölder continuity of $\|\cdot\|^p$ [RN2020]:

$$\|v + w\|^p \leq \|v\|^p + p \frac{\langle v, w \rangle}{\|v\|^{2-p}} + 2^{2-p} \|w\|^p.$$

High Probability Lower Bound

Theorem 2. For any algorithm satisfying the cone condition,

$$x_T^{\text{out}} \in x_1 - \text{Cone} \{ \nabla f(x_1, \xi_1), \dots, \nabla f(x_{T-1}, \xi_{T-1}) \},$$

there exists a convex problem with **A.1.**: for any $\alpha > p$, w. p. δ

$$F(x_T^{\text{out}}) - F(x^*) \geq \frac{\|x_1 - x^*\|}{2(T-1)^{\frac{\alpha-1}{\alpha}}} \left(\frac{1}{4\alpha\delta} \right)^{1/\alpha}.$$

Implication: Any non-adaptive algorithm, e.g., SGD, suffers from polynomial dependence in $1/\delta$; different from $\log(1/\delta)$ for Clip-SGD.

References

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Lower Bound Construction

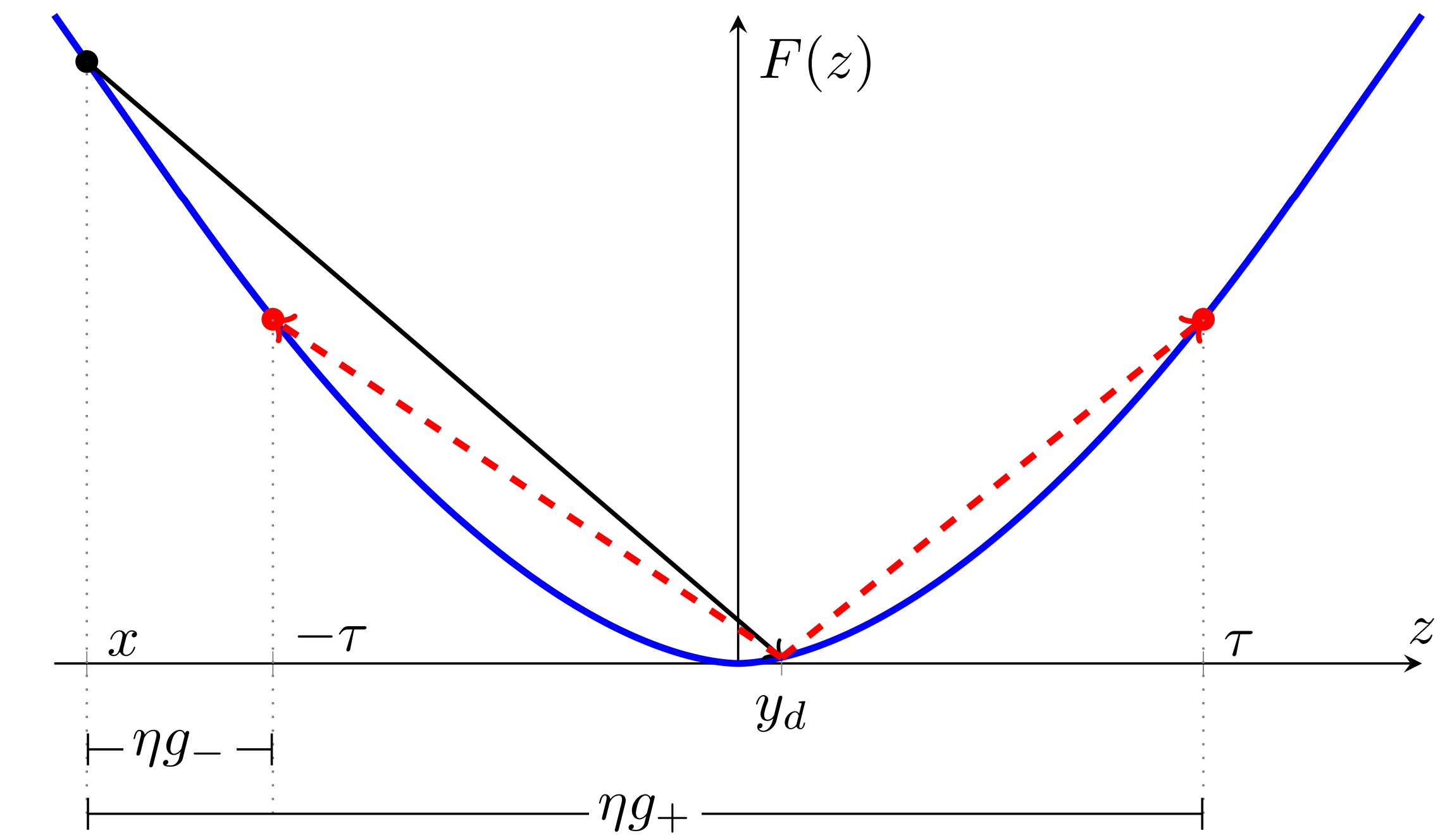


Figure 2: Visualisation of the lower bound construction in Theorem 1.3., for the case where the deterministic update $y_d = x - \eta \nabla F(x)$ lands within the critical radius τ . The dashed lines represent the two possible offsets introduced by the constructed gradient oracle, resulting in $y = x - \eta g_{\pm} u = x + \eta g_{\pm} = \pm \tau$.

Experiments

Problem: $F(x) := \|Ax - b\|_1 + \frac{\mu}{2} \|x\|_2^2$, $\mathcal{X} := \{x \in \mathbb{R}^d : \|x\|_{\infty} \leq R\}$,

Gradient noise: $\nabla f(x, \xi) := \nabla F(x) + \xi$ with ξ is a two-sided α -Pareto.

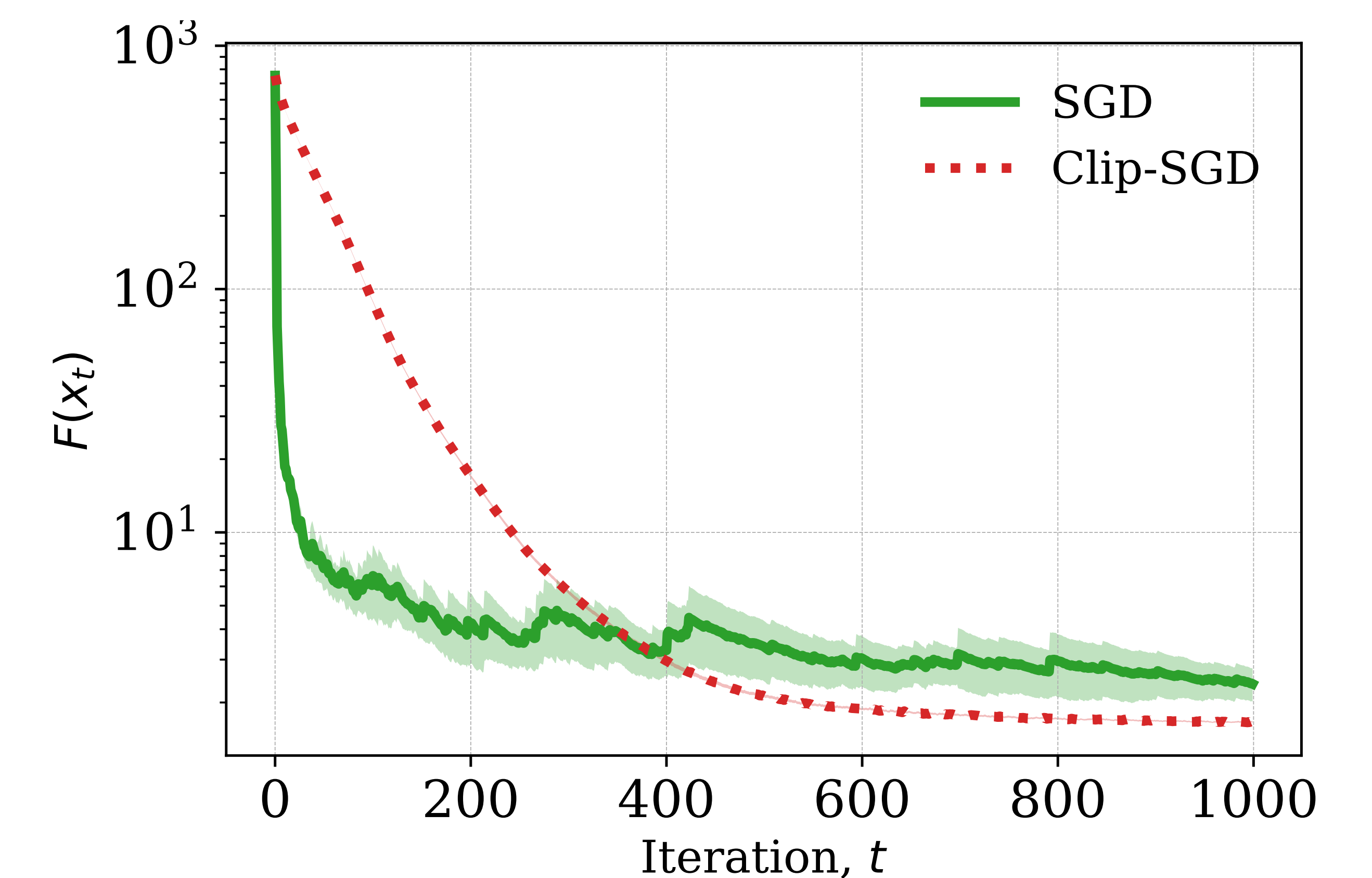


Figure 3: SGD vs. Clip-SGD for $\alpha = 1.6$.

$$\text{Clip-SGD: } x_{t+1} = \Pi_{\mathcal{X}}(x_t - \eta_t g_t), \quad g_t = \text{clip}(\nabla f(x_t, \xi_t), \lambda_t)$$

with $\text{clip}(v, \lambda) := v \cdot \min\{1, \lambda/\|v\|_2\}$. Clipping schedule: $\lambda_t = \lambda_1 t^{1/p}$.

Key insights:

- ✓ SGD automatically adapts to p and doesn't stall optimization.
- ✓ SGD can be faster than Clip-SGD if clipping λ_1 is untuned.